# Anisah Rihadatul Aisy

**Quality Control | Chemical Analyst | Research & Development**
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## PROFESSIONAL SUMMARY

Chemistry graduate from Brawijaya University (GPA 3.60/4.00) with hands-on experience in laboratory analyst, Quality Control, Research & Development. Experienced in water quality testing, sample preparation, extraction, UV-Vis spectrophotometry, FTIR, and formulation research, including Nanostructured Lipid Carrier (NLC) preparation, antioxidant analysis, microscopy, and in-vitro permeation studies. Also experienced in photophysical characterization and fluorescence lifetime. Trained in GLP, GMP/CPOB, and HACCP documentation. Accurate in data recording, interpretation, and scientific reporting.

## EDUCATION

**Bachelor of Science in Chemistry** | Brawijaya University, Malang | August 2022 – July 2026
GPA: 3.60/4.00

**Thesis:** A Study of the Antioxidant Potential and Permeation Rate of NLC-Cur-CTAB in Natural Membranes

- Developed NLC-Cur-CTAB using the PIT method and formulated it into a Carbopol-based gel, achieving 133 nm particle size and 95.57% drug loading.
- Evaluated antioxidant activity using the DPPH method, obtaining an IC50 of 75.44 µg/mL.
- Analyzed in-vitro permeation using a Franz diffusion cell, showing an approximately 22-fold increase in permeation rate during the evaluated period.

## LABORATORY & RESEARCH EXPERIENCE

**Laboratory Assistant, Biochemistry** | Brawijaya University | February – May 2026

- Supervised 6 biochemistry practical sessions for 28 undergraduate students in an English-instructed class.
- Guided qualitative and quantitative analyses, including carbohydrate testing, amino acid and protein identification, lipid and vitamin analysis, and enzyme activity assays.
- Delivered technical explanations, monitored laboratory procedures and reinforced students' understanding of biochemical analysis and laboratory safety.

**Chemical Analyst Intern** | Forensic Drug Laboratory, Polda Jatim | January 2026

- Screened 50+ forensic samples per day using Marquis-Simon tests and TruNarc.
- Analyzed suspected methamphetamine, cannabis, and drug tablets.
- Observed GC-MS analysis and compiled findings.

**Laboratory Assistant, Organic Chemistry** | Brawijaya University | August – December 2025

- Assisted 42 undergraduate students in Organic Chemistry laboratory sessions by guiding experimental procedures and explaining fundamental chemical principles.
- Supervised 4 experiments: caffeine isolation from tea leaves, curcuminoid extraction from turmeric, trimyristin hydrolysis from nutmeg, and compound identification in ginger essential oil.
- Supported FTIR and UV-Vis analysis and assisted in interpreting experimental results.

**Research Assistant** | Physical Chemistry Laboratory, Brawijaya University | August – December 2025

- Formulated curcumin-loaded NLC using the PIT method, producing homogeneous particles with an average size of 326 nm.
- Isolated curcuminoids by Soxhlet extraction (9.44%–18% yield) and identified them by LC-HRMS (98.3% compound matching).
- Evaluated photophysical stability across four solvent polarities and basic pH using UV-Vis, spectrofluorometer, and CLSM; documented findings for a SINTA 2 journal manuscript.

**Quality Control Intern** | PDAM Delta Tirta Sidoarjo | June – August 2025

- Tested water quality in Buduran and Krian using a pH meter, turbidimeter, and spectrophotometer, and recorded the results.
- Performed jar tests at 50–140 ppm PAC, evaluating coagulation performance by turbidity, pH, and color.
- Determined 90 ppm PAC as the optimum dose for Buduran, achieving turbidity of 2.71 ± 0.5 NTU

**Student Exchange Program** | Universiti Malaysia Terengganu, Malaysia | April – May 2025

- Collected water samples at 0, 5, 10, 15, and 20 m depths using a 5 L Niskin sampler.
- Determined dissolved oxygen by Winkler titration using 0.0125 M $Na_2S_2O_3$.
- Analyzed phosphorus by UV-Vis spectrophotometry using duplicate blanks.

## ORGANIZATIONAL EXPERIENCE

**Vice Head, Qur'anic Scientific Writing & App Design** | Brawijaya University | February – December 2025

- Coordinated a 5-member team and weekly mentoring sessions.

- Prepared and delivered mentoring materials and monitored members' progress.

## SKILLS

**Laboratory & Quality Control:** Sampling, Sample Preparation, Qualitative & Quantitative Analysis, Titration, Soxhlet Extraction, Data Recording & Reporting, Laboratory Testing, Product Evaluation, Antioxidant Analysis, In-Vitro Permeation Study, Microscopy

**Analytical Instruments:** UV-Vis Spectrophotometer (Intermediate), FTIR (Basic–Intermediate), GC-MS (Basic), TruNarc Analyzer (Experienced), HPLC (Basic), pH Meter, Turbidimeter

**Quality & Safety Practices:** Good Laboratory Practice (GLP), Good Manufacturing Practice (GMP/CPOB), HACCP Documentation, Laboratory Safety, Quality Control Procedures, Data Accuracy

**Research & Formulation:** Experimental Work, Research Methodology, Literature Review, Scientific Writing, Nanostructured Lipid Carrier (NLC) Preparation, Nanoemulsion, Photophysical Analysis

## TRAINING & CERTIFICATIONS

- Healthcare Regulatory – Drugs & Cosmetics, Ioda Academy (September 2026): regulatory readiness assessment and 3 applied projects on documentation gaps, labeling and claims compliance under BPOM requirements.
- HACCP Documentation Development Training, Batch 32, Makin Ahli (May 2026): hazard analysis, CCP determination, monitoring, corrective actions, validation, and verification.
- Good Manufacturing Practice (GMP/CPOB) for Pharmaceutical Training, Batch 12, Makin Ahli (February 2026): quality assurance, documentation, hygiene, sanitation, production procedures, and quality control.
- Research & Development Specialist – Flavour Industry, Latih (December 2025): R&D workflow and cross-functional collaboration with QC, QA, Regulatory, and Production.
- Online Mini Bootcamp: Food Laboratory, Mindo Education (September 2025): food analysis, Good Laboratory Practice (GLP), laboratory safety, food microbiology, and introduction to HPLC.

## ACHIEVEMENTS

- Best Student, Student Mobility Program, Universiti Malaysia Terengganu (2025)
- Gold Medal, National Scientific Writing Competition, NASPO (2024)
- Silver Medal, International Scientific Writing Competition, WYIIA (2024)
- 2nd Place, Futuristic Science Idea of The Quran, Brawijaya University (2024)
- Awardee, Beasiswa Pendidikan Indonesia, Ministry of Education and Culture (2022)
- Silver Medal, National Scientific Writing Competition, NASPO (2021)